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UNIT 1

Exp 1: Average of Three Numbers

Code:

```
#include <stdio.h>

int main() {
    float a, b, c, average;

    printf("Enter three numbers: ");
    scanf("%f %f %f", &a, &b, &c);

    average = (a + b + c) / 3;

    printf("Average of the three numbers = %.2f\n", average);

    return 0;
}
```

Output:

```
C:\Users\Admin\Desktop\C_Codes> gcc exp1.c -o exp1
C:\Users\Admin\Desktop\C_Codes> exp1
Enter three numbers: 12 18 24
Average of the three numbers = 18.00
C:\Users\Admin\Desktop\C_Codes>
```

Exp 2: Swap Two Numbers Using Temporary Variable

Code:

```
#include <stdio.h>

int main() {
    int x, y, temp;

    printf("Enter value of x: ");
    scanf("%d", &x);
    printf("Enter value of y: ");
    scanf("%d", &y);

    temp = x;
    x = y;
    y = temp;

    printf("Values after swapping:\n");
    printf("x = %d, y = %d\n", x, y);

    return 0;
}
```

Output:

```
C:\Users\Admin\Desktop\C_Codes> gcc exp2.c -o exp2
C:\Users\Admin\Desktop\C_Codes> exp2
Enter value of x: 10
Enter value of y: 90
Values after swapping:
x = 90, y = 10
C:\Users\Admin\Desktop\C_Codes>
```

Exp 3: Area of Circle

Code:

```
#include <stdio.h>
#define PI 3.14159

int main() {
    float r, area;

    printf("Enter radius: ");
    scanf("%f", &r);

    area = PI * r * r;

    printf("Area = %.2f\n", area);

    return 0;
}
```

Output:

```
C:\Users\Admin\Desktop\C_Codes> gcc exp3.c -o exp3
C:\Users\Admin\Desktop\C_Codes> exp3
Enter radius: 4.5
Area = 63.62
C:\Users\Admin\Desktop\C_Codes>
```

UNIT 2

Exp 4: Positive, Negative, or Zero

Code:

```
#include <stdio.h>

int main() {
    int n;

    printf("Enter any integer: ");
    scanf("%d", &n);

    if (n > 0) {
        printf("Result: %d is Positive\n", n);
    } else if (n < 0) {
        printf("Result: %d is Negative\n", n);
    } else {
        printf("Result: Entered number is Zero\n");
    }

    return 0;
}
```

Output:

```
C:\Users\Admin\Desktop\C_Codes> gcc exp4.c -o exp4
C:\Users\Admin\Desktop\C_Codes> exp4
Enter any integer: -45
Result: -45 is Negative
C:\Users\Admin\Desktop\C_Codes>
```

Exp 5: Check Even or Odd

Code:

```c
#include <stdio.h>

int main() {
    int val;

    printf("Enter number: ");
    scanf("%d", &val);

    if (val % 2 == 0)
        printf("%d is Even number\n", val);
    else
        printf("%d is Odd number\n", val);

    return 0;
}
```

Output:

```
C:\Users\Admin\Desktop\C_Codes> gcc exp5.c -o exp5
C:\Users\Admin\Desktop\C_Codes> exp5
Enter number: 57
57 is Odd number
C:\Users\Admin\Desktop\C_Codes>
```

Exp 6: Multiplication Table (Loops)

Code:

```c
#include <stdio.h>

int main() {
    int num, i;

    printf("Enter number for table: ");
    scanf("%d", &num);

    printf("Table using for loop:\n");
    for (i = 1; i <= 10; i++) {
        printf("%d x %d = %d\n", num, i, num * i);
    }

    return 0;
}
```

Output:

```
C:\Users\Admin\Desktop\C_Codes> gcc exp6.c -o exp6
C:\Users\Admin\Desktop\C_Codes> exp6
Enter number for table: 8
Table using for loop:
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40
8 x 6 = 48
8 x 7 = 56
8 x 8 = 64
8 x 9 = 72
8 x 10 = 80
C:\Users\Admin\Desktop\C_Codes>
```

UNIT 3

Exp 7: Factorial using Function

Code:

```
#include <stdio.h>

long calcFactorial(int n) {
    long f = 1;
    for (int i = 1; i <= n; i++) {
        f *= i;
    }
    return f;
}

int main() {
    int n;
    printf("Enter a positive number: ");
    scanf("%d", &n);

    printf("Factorial of %d = %ld\n", n, calcFactorial(n));

    return 0;
}
```

Output:

```
C:\Users\Admin\Desktop\C_Codes> gcc exp7.c -o exp7
C:\Users\Admin\Desktop\C_Codes> exp7
Enter a positive number: 6
Factorial of 6 = 720
C:\Users\Admin\Desktop\C_Codes>
```

Exp 8: Sum of 1D Array Elements

Code:

```c
#include <stdio.h>

int main() {
    int a[50], n, total = 0;

    printf("Enter array size: ");
    scanf("%d", &n);

    printf("Enter %d values: ", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
        total += a[i];
    }

    printf("Total sum = %d\n", total);

    return 0;
}
```

Output:

```
C:\Users\Admin\Desktop\C_Codes> gcc exp8.c -o exp8
C:\Users\Admin\Desktop\C_Codes> exp8
Enter array size: 4
Enter 4 values: 15 25 35 45
Total sum = 120
C:\Users\Admin\Desktop\C_Codes>
```

Exp 9: Matrix Addition (2D Array)

Code:

```
#include <stdio.h>

int main() {
    int m1[2][2], m2[2][2], sum[2][2];

    printf("Enter 4 elements of 1st 2x2 matrix:\n");
    for (int i = 0; i < 2; i++)
        for (int j = 0; j < 2; j++)
            scanf("%d", &m1[i][j]);

    printf("Enter 4 elements of 2nd 2x2 matrix:\n");
    for (int i = 0; i < 2; i++)
        for (int j = 0; j < 2; j++)
            scanf("%d", &m2[i][j]);

    printf("Sum Matrix:\n");
    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 2; j++) {
            sum[i][j] = m1[i][j] + m2[i][j];
            printf("%d ", sum[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

Output:

```
C:\Users\Admin\Desktop\C_Codes> gcc exp9.c -o exp9
C:\Users\Admin\Desktop\C_Codes> exp9
Enter 4 elements of 1st 2x2 matrix:
2 4 6 8
Enter 4 elements of 2nd 2x2 matrix:
1 3 5 7
Sum Matrix:
3 7
11 15
C:\Users\Admin\Desktop\C_Codes>
```